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MS AZURE TUTORIAL

Special Topic in Data Engineering (SECP3843)

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Project Overview

The present visibility gap of the company in relation to customer demographics is a critical one that concerns the effects of gender distribution on the purchasing behaviour of various product categories. Although a considerable amount of data is stored in the on premises SQL database, because the information is not used and is siloed which leads to the leadership unable to make decisions based on data about marketing and inventory.

This project aims to fill in the gap by moving on premises data to a scalable Azure cloud infrastructure and developing an automated daily data pipeline that can be sustained independently. Through centralization of these records, the business will create an all inclusive KPI Dashboard to give real time data on total sales revenue, product volume and demographic splits. This change is a competitive imperative in the contemporary retail environment with awareness of these trends providing the ability to market to targeted customers and better customer retention. The project scope is well stipulated as end-to-end engineering of a data lakehouse with the AdventureWorks dataset, including ingestion via clouds, multi-stage transformation with a Medallion Architecture and eventual deployment of an interactive Power BI interface.

Description of The Solution

To support these business needs, a strong Cloud Data Lakehouse was developed on Microsoft Azure with the Medallion Design Pattern to maintain data quality and lineage with Bronze, Silver, and Gold layers.

The logical workflow starts with Azure Data Factory and a Self-Hosted Integration runtime that safely retrieves the raw data in the local environment and loads it in the Azure Data Lake storage Gen2 in Parquet files as the Bronze layer. After that, the Azure Databricks is used to clean and standardize data in the Silver layer, messy timestamps are converted and null values are managed and the result is saved in Delta format in order to perform ACID transactions. In the final Gold layer, a secondary Databricks process applies refined business logic and renames columns to a user-friendly format, creating a reporting-ready dataset that is served via Azure Synapse Analytics serverless SQL views.

This particular design option enables to perform highly without the high costs of a dedicated SQL pool since only costs during active usage. These views are then fed into the final Power BI dashboard giving stakeholders an automated, daily-refreshed interface that allows date-based queries and demographic filtering options so that strategic decisions of the company are always made using the most accurate and up to date available data.

Technology Use (Azure Services)

The implementation of data lake warehouse system utilized a series of integrated Azure services, each of which played a specific role throughout the entire end-to-end process. Firstly, Azure SQL Database served as the source environment, used to host the local AdventureWorks dataset. Besides that, Azure Data Factory (ADF) was responsible for the entire workflow, using self-hosted integration runtime to securely connect the local SQL environment with cloud storage. Furthermore, Azure Data Lake Storage (ADLS) Gen2 provided a central storage foundation for the Medallion architecture, accommodating the partitioned bronze, silver and gold layers.

Azure Databricks served as the main computing engine; it was used to clean and standardize the raw data in the silver layer and apply fine business logic using PySpark in the gold layer. To provide a high-performance, cost-effective consumption layer, Azure Synapse Analytics (Serverless SQL pool) was used to create SQL views on the Delta files in the gold layer. Finally, MS Power BI provided a visual interface, directly connected to Synapse to provide real-time data on key performance indicators such as sales revenue and customer demographics, to help the leadership make data-based decisions.

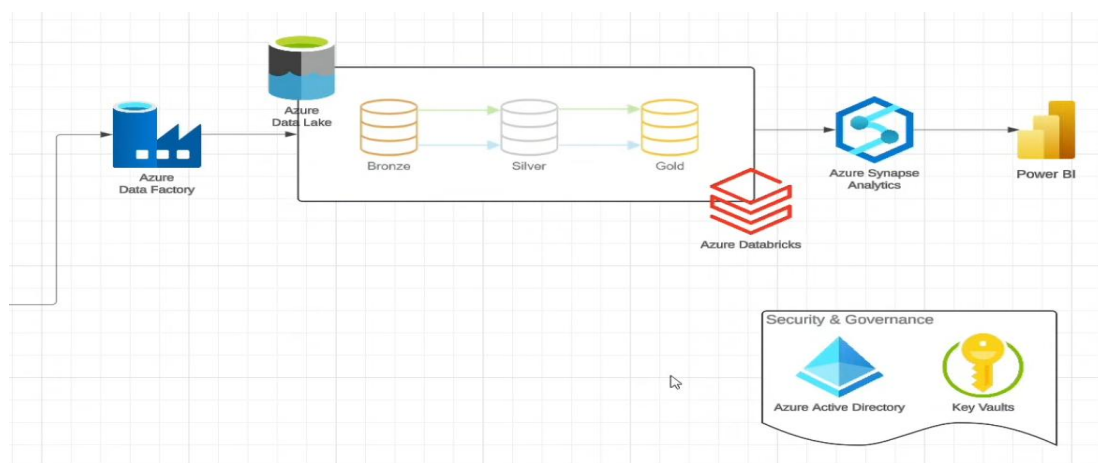


Figure 1: End-to-End Azure Data Engineering Architecture showing the integration of Azure Data Factory, Data Lake Storage Gen2 (Bronze, Silver, Gold), Databricks, Synapse Analytics, and Power BI.

Data Pipeline Architecture

This architecture follows the modern Medallion Design Pattern to maintain high data quality and historical records throughout the data ingestion and transformation stages.

In the Bronze (Ingestion) stage, raw data is extracted from the local SQL database using ADF and stored in ADLS Gen2 in Parquet format. Next, during the Silver (Standardization) stage, Databricks will convert these raw files into Delta format, and handle data cleaning tasks such as converting messy timestamps and managing null values. The reason is to make sure that the data is available for ACID transactions. Lastly, The Gold (Refinement) stage involves a secondary Databricks process that applies business transformations and renames columns to a user-friendly, reportable format.

For the Consumption (analysis) layer, Azure Synapse Analytics creates serverless views based on the gold Delta files. This specific design allows the system to operate efficiently without incurring the high costs associated with dedicated SQL pools, as it only incurs expenses when conducting active queries through Power BI.

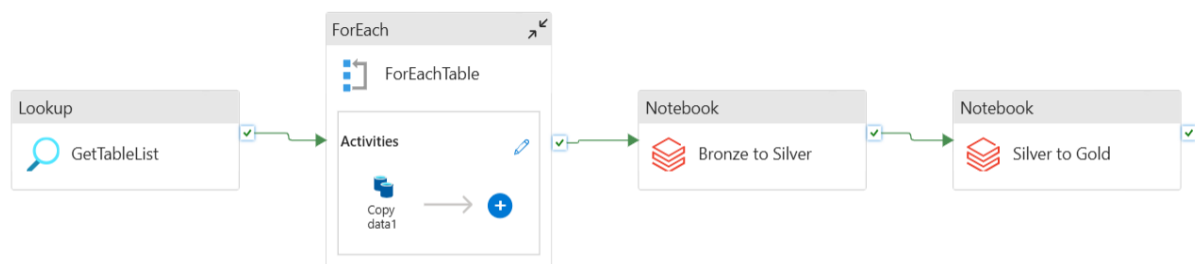
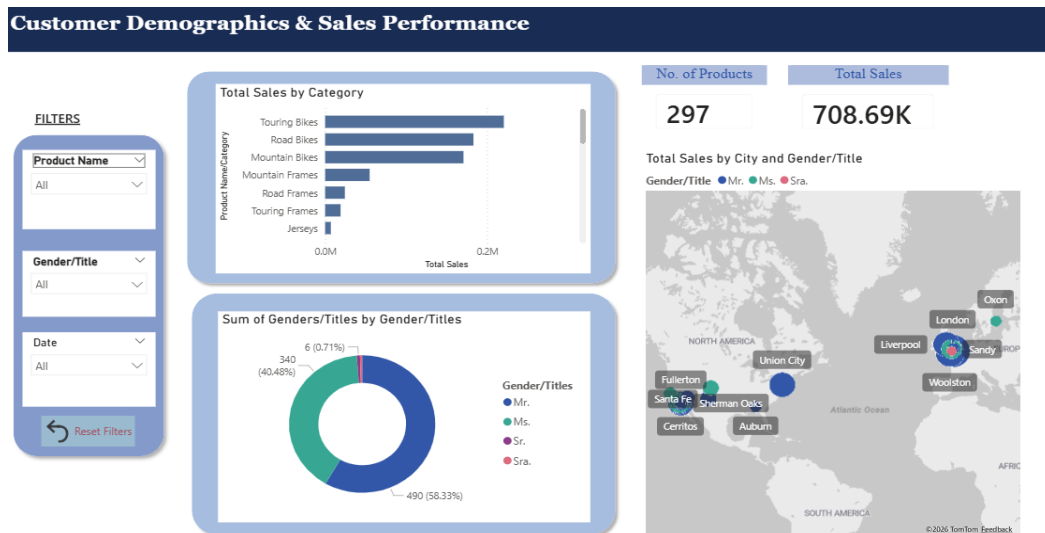


Figure 2: End-to-End Data Pipeline showing ingestion from SQL Server to Databricks transformation

MS Power BI Dashboard

The final stage of the project is the deployment of an interactive MS Power BI dashboard that is designed with a modern web app interface to bridge the company's customer information gap.



In the dashboard, it has a high level KPI cards detailing total sales revenue (708.69K) and product volume (297) on a single screen. To support strategic decision-making, it integrates a Geographic Map to visualize sales by city and a Donut Chart representing gender distribution, revealing that 58% of the customer base identifies as "Mr."

The solution offers a dedicated Filter Sidebar where stakeholders can slice data by Product, Gender, and Date to get particular demographics with ease. The dashboard allows the stakeholders to constantly have access to the correct and current information by using the automated daily pipeline. This change eliminates the old silos of information and allows proactive inventory control. It also gives the leadership team a real-time, daily updated interface that shows the health of the business.

Reflections

1. Brendan Chia Yan Fei

This project taught me how to troubleshoot different problems in a data pipeline. One thing that is different from my expectation is configuring the secret. While normally we would just enter the name and password, here we need to use Azure Key Vault to store the secret, and we would just implement Azure Key Vault for our different Services. Besides, since I use MS Azure to do a project before, I care a lot about the cost usage. Hence, I researched on how to minimise our cost for this data pipeline, including disabling the data factory triggers, and reducing the time for auto termination for my Databricks compute.

2. Chew Chiu Xian

Completing this end-to-end project based on Azure was a very meaningful experience, connecting the theoretical data engineering with the significant gap between industry-standard practices. One of the biggest challenges I faced was managing Delta transaction logs and resolving pattern mismatch issues in Synapse Analytics. I learned that data types must be strictly converted, especially during the transformation from silver level to gold level to avoid problems such as "boolean/false" errors that occur during visualization.

Through collaboration with Azure Data Factory, I realized the importance of automation and how self-hosted integration runtime serves as a secure bridge between local data and the cloud. Additionally, implementing the "badge architecture" helped me understand how to maintain data integrity at different refinement stages. This project greatly enhanced my technical skills in PySpark and SQL, and highlighted the importance of cost-effective design, such as using Synapse Serverless Pools. Overall, I now have greater confidence in my ability to build and deploy fully functional, automated cloud data pipelines.

3. Ng Yu Hin

This Azure Tutorial lesson has demonstrated how a data pipeline architecture looks like, by using tools such as Microsoft Azure's Data Factory and DataBricks, as well as Power BI for its interactive dashboard. Building this end to end Azure pipeline was a massive eye-opener, for the most part it wasn't actually the coding but the environment configuration. A lot of time was spent during troubleshooting the Databricks cluster settings to find a balance between performance and the limitations of a trial account. It taught me that cloud engineering is just as much about managing resources as it is about moving data.

The most frustrating part was the Azure Data Factory connectivity where the tutorial was a bit vague. Through a lot of trial and error and brainstorming, the link between the local SQL and the cloud was able to be stable. Implementing the Medallion Architecture cleared up how to maintain data integrity seeing the data transform from raw Parquet in Bronze to clean, renamed Delta tables in Gold was insightful. This project definitely boosted my confidence in PySpark and SQL, and it showed me why cost effective tools like Synapse Serverless Pools are so important in the industry. In the end, I got to build an automated pipeline that doesn't just work on paper, but actually stays running in production.